EBS Volume Management and Backup Strategies

What is EBS volume

An EBS (Elastic Block Store) volume is a persistent block-level storage device which is provided by Amazon Web Services that can be attached to an EC2 instance to store data, similar to a virtual hard disk.

TWO TYPES OF BLOCK STORAGE DEVICES FOR EC2

1. EBS Volume
2. Instance Store

1. EBS Volume (EBS backed EC2 instance)

1. EBS volume is a persistent block level storage which is provided by amazon web service that u can attach to your ec2 instance
2. EBS volume behaves like ram, Unformatted, external block storage devices that you can attach to your EC2 instance.
3. EBS volume stores the data in the form of blocks we known as data chunks that would not contain the metadata it only contains the address for each block
4. EBS volumes are block storage devices suitable for database style data that requires frequent reads and writes.
5. EBS volumes are attached to your EC2 instances through the AWS network like virtual hard disk.
6. an EBS volume is availability zone specific.
7. Both EBS volume and EC2 instance must be in the same AZ.
8. It is a persistent block level storage means if u have to stop or reboot the EC2 instance the data will not be deleted. but if you terminate the instance the data will be deleted. (by default)
9. But when we remove the delete on termination tick, the EBS volume should not be deleted even after deleting/terminating the EC2 instance.
10. It is a network attached storage so its slow storage compared to instance storage.
11. We can create up to 5000 EBS volumes per region

2. Instance store (instance store backed EC2 instance)

1. Instance store backed EC2 basically directly attached to our ec2 instance.
2. It is limited to 10GB per device.

3. It is a Ephemeral storage means non persistent storage so if u terminate the EC2 instance the data will be deleted. but suppose if u rebooted the instance the data will not be deleted.
4. The EC2 instance can't be stopped, can only be rebooted or terminated.
5. And its a direct attach storage so it is faster than EBS volume.

TYPES OF EBS VOLUME

1.SSD Backed Volume (Solid State Drive)

These are bootable volumes means we can install OS on these drive

1. General Purpose (Gp2, Gp3)
2. Provisional IOPS (Io2 Block Express, Io1, Io2)

2.HHD Backed Volume (Hard Disk Drive)

These are non-bootable volumes means we cannot install OS On this drive because we use this volume as a storage

1. Throughput Optimized HDD (St1)
2. Cold HDD (Sc1)

3.Magnetic Standard

These are bootable volumes means we can install OS on these drive

Note:

1.Throughput

1. amount of data transferred per second (mb/s)
2. Example:
   If 100 MB of data is transferred in 1 second → Throughput = 100 MB/s
3. Used for:
   - Large file transfers
   - Streaming
   - Backup operations

2.IOPS (Input Output Per Second)

1. number of read/write operations per second
2. Example:
   If a disk performs 500 read/write operations in 1 second → IOPS = 500
3. Used for:
   - Databases
   - Transaction-heavy apps
   - Random access workloads

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	General Purpose SSD volumes		Provisioned IOPS SSD volumes	
Volume type	gp3	gp2	io2 Block Express [3]	io1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)		99.999% durability (0.001% annual failure rate)	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)
Use cases	• Transactional workloads • Virtual desktops • Medium-sized, single-instance databases • Low-latency interactive applications • Boot volumes • Development and test environments		Workloads that require: • Sub-millisecond latency • Sustained IOPS performance • More than 64,000 IOPS or 1,000 MiB/s of throughput	• Workloads that require sustained IOPS performance or more than 16,000 IOPS • I/O-intensive database workloads
Volume size	1 GiB - 16 TiB		4 GiB - 64 TiB [4]	4 GiB - 16 TiB
Max IOPS per volume	16,000 (64 KiB I/O)	16,000 (16 KiB I/O)	256,000 (16 KiB I/O) [5]	64,000 (16 KiB I/O)
Max throughput per volume	1,000 MiB/s	250 MiB/s [1]	4,000 MiB/s	1,000 MiB/s [2]
Amazon EBS Multi-attach	Not supported		Supported	
NVMe reservations	Not supported		Supported	Not supported
Boot volume	Supported			

	Throughput Optimized HDD volumes	Cold HDD volumes
Volume type	st1	sc1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)	
Use cases	• Big data • Data warehouses • Log processing	• Throughput-oriented storage for data that is infrequently accessed • Scenarios where the lowest storage cost is important
Volume size	125 GiB - 16 TiB	
Max IOPS per volume (1 MiB I/O)	500	250
Max throughput per volume	500 MiB/s	250 MiB/s
Amazon EBS Multi-attach	Not supported	
Boot volume	Not supported	

	Magnetic
Volume type	standard
Use cases	Workloads where data is infrequently accessed
Volume size	1 GiB-1 TiB
Max IOPS per volume	40–200
Max throughput per volume	40–90 MiB/s
Boot volume	Supported

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